Fall 2025 EE582 Homework 02 Report
Nodal-Analysis-based Modeling of Electric Circuits

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Due: Tuesday, October 7, 2025

Contents

Section A (Numerical Oscillations of Discretization Rules): Question 1

One way to analyze the properties of various discretization methods (e.g., in terms of their numerical oscillations) is to solve a lossless first-order system (e.g., inductance or capacitance) when the state variable of the component is suddenly changed. The circuit on the right considers a case where a dc voltage is suddenly applied to a capacitor. Because the source voltage is applied like a step function, we know that the current of the capacitor should look like an impulse, analytically. Let us assume that in the simulations all the variables are zero at $t = 0$ and the voltage source takes its value at $t = \Delta t$.

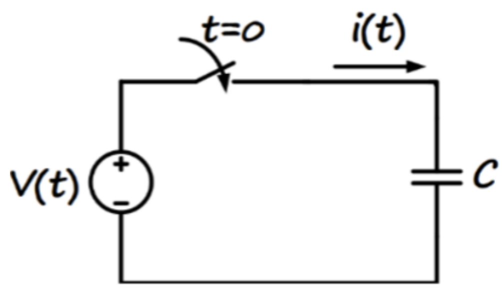

Figure 1: Simple capacitor circuit with DC voltage source

1) Determine the step-by-step solution for the current in the capacitor (in terms of capacitance C, time-step size Δt, and source voltage) using the following discretization rules: a) Forward Euler, b) Backward Euler, c) Trapezoidal.

Note: I'm assuming that the capacitor has an initial voltage V_0 at $t = 0^-$ for generating the state variable trajectory tables. However in question 2, I simulate the case with $V_0 = 0$ V.

A.1.1 Trapezoidal Discretization of Capacitor

The equations for the Trapezoidal discretization of a capacitor are (see Appendix I for derivation):

The final equations for $v(t)$ and $i(t)$ are as follows:

$$\boxed{v(t) = e_h + R_C i(t)} \tag{1}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{2}$$

where:

- $R_C = \frac{\Delta t}{2C}$ is the equivalent resistance.

- $e_h = v(t - \Delta t) + R_C i(t - \Delta t)$ is the history voltage.
- $i_h = \frac{v(t-\Delta t)}{R_C} - i(t - \Delta t)$ is the history current.

Step-by-Step Solution Table for Trapezoidal Discretization

Table 1 shows the step-by-step evolution of the system variables based on trapezoidal discretization.

t	$v(t)$	$i(t)$	e_h	i_h
-h	V_0	0	–	–
0^+	V_{DC}	$\frac{V_{DC}}{R_C} - \frac{V_0}{R_C}$	V_0	$\frac{V_0}{R_C}$
h	V_{DC}	$-\frac{V_{DC}}{R_C} + \frac{V_0}{R_C}$	$2V_{DC} - V_0$	$\frac{2V_{DC}}{R_C} - \frac{V_0}{R_C}$
$2h$	V_{DC}	$\frac{V_{DC}}{R_C} - \frac{V_0}{R_C}$	V_0	$\frac{V_0}{R_C}$
$3h$	V_{DC}	$-\frac{V_{DC}}{R_C} + \frac{V_0}{R_C}$	$2V_{DC} - V_0$	$\frac{2V_{DC}}{R_C} - \frac{V_0}{R_C}$
$4h$	V_{DC}	$\frac{V_{DC}}{R_C} - \frac{V_0}{R_C}$	V_0	$\frac{V_0}{R_C}$

Table 1: Step-by-step solution for trapezoidal discretization of circuit in Figure 1

A.1.2 Backward-Euler (BE) Discretization of Capacitor

The equations for the Backward-Euler discretization of a capacitor are (see Appendix II for derivation):

$$\boxed{v(t) = e_h + R_C\, i(t)} \tag{3}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{4}$$

where:

- $R_C = \dfrac{\Delta t}{C}$
- $e_h = v(t - \Delta t) + R_C i(t - \Delta t)$
- $i_h = \dfrac{v(t - \Delta t)}{R_C}$

Step-by-Step Solution Table for Backward-Euler Discretization

Table 2 shows the step-by-step evolution of the system variables based on backward euler discretization.

t	$v(t)$	$i(t)$	e_h	i_h
$-h$	V_0	0	$-$	$-$
0^+	V_{DC}	$\frac{V_{DC}}{R_C} - \frac{V_0}{R_C}$	V_0	$\frac{V_0}{R_C}$
h	V_{DC}	0	V_{DC}	$\frac{V_0}{R_C}$
$2h$	V_{DC}	0	V_{DC}	$\frac{V_{DC}}{R_C}$
$3h$	V_{DC}	0	V_{DC}	$\frac{V_{DC}}{R_C}$
$4h$	V_{DC}	0	V_{DC}	$\frac{V_{DC}}{R_C}$

Table 2: Step-by-step solution for backward euler discretization of circuit in Figure 1

A.1.3 Forward Euler (FE) Discretization of Capacitor

Upon solving for the forward euler discretization of the capacitor, we encounter the issue of division by zero as $v(t)$ and $e_h(t)$ are equal at every time step (one source connected to another source directly), and $i(t)$ becomes undefined (because of lack of any resistance between them). $i(t)$ could be of an arbitrarily high magnitude, however in order to ascertain its sign, we introduce a very small resistance $R_\epsilon \ll R_c$ in series with the capacitor to avoid division by zero. This resistance is artificial and does not represent any physical element in the circuit (or at best could be argued to represent the relatively negligible resistance of the connecting wires).

The equations for the Forward Euler discretization of a capacitor are (see Appendix III for derivation):

$$\boxed{v(t) = e_h(t)} \tag{5}$$

$$\boxed{i(t) = \frac{v(t)}{R_\epsilon} - i_h(t)} \tag{6}$$

where:

- $R_c = \dfrac{h}{C}$ (discretized capacitor resistance)
- $R_\epsilon \ll \dfrac{h}{C}$ (artificial resistance for FE table)
- $e_h(t) = v(t-h) + R_c i(t-h)$
- $i_h(t) = \frac{e_h(t)}{R_\epsilon}$

Step-by-Step Solution Table for Forward-Euler Discretization

Table 3 shows the step-by-step evolution of the system variables based on forward euler discretization, assuming a small resistance R_ϵ to avoid division by zero.

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t	$v(t)$	$i(t)$	$e_h(t)$	$i_h(t)$
$-h$	V_0	0	$-$	$-$
0^+	V_{DC}	$\dfrac{V_{DC}-V_0}{R_\epsilon}$	V_0	$\dfrac{V_0}{R_\epsilon}$
h	V_{DC}	$\dfrac{V_0-V_{DC}}{R_\epsilon^2/R_c}$	$V_{DC}\left(1+\frac{R_c}{R_\epsilon}\right)-V_0\left(\frac{R_c}{R_\epsilon}\right)$	$\dfrac{V_{DC}\left(1+\frac{R_c}{R_\epsilon}\right)-V_0\left(\frac{R_c}{R_\epsilon}\right)}{R_\epsilon}$
$2h$	V_{DC}	$\dfrac{V_{DC}-V_0}{R_\epsilon^3/R_c^2}$	$-V_{DC}\left(\frac{R_c^2}{R_\epsilon^2}-1\right)+V_0\left(\frac{R_c^2}{R_\epsilon^2}\right)$	$\dfrac{-V_{DC}\left(\frac{R_c^2}{R_\epsilon^2}-1\right)+V_0\left(\frac{R_c^2}{R_\epsilon^2}\right)}{R_\epsilon}$
$3h$	V_{DC}	$\dfrac{V_0-V_{DC}}{R_\epsilon^4/R_c^3}$	$V_{DC}\left(1+\frac{R_c^3}{R_\epsilon^3}\right)-V_0\left(\frac{R_c^3}{R_\epsilon^3}\right)$	$\dfrac{V_{DC}\left(1+\frac{R_c^3}{R_\epsilon^3}\right)-V_0\left(\frac{R_c^3}{R_\epsilon^3}\right)}{R_\epsilon}$
$4h$	V_{DC}	$\dfrac{V_{DC}-V_0}{R_\epsilon^5/R_c^4}$	$-V_{DC}\left(\frac{R_c^4}{R_\epsilon^4}-1\right)+V_0\left(\frac{R_c^4}{R_\epsilon^4}\right)$	$\dfrac{-V_{DC}\left(\frac{R_c^4}{R_\epsilon^4}-1\right)+V_0\left(\frac{R_c^4}{R_\epsilon^4}\right)}{R_\epsilon}$

Table 3: Step-by-step solution for forward euler discretization of circuit in Figure 1. Note the diverging oscillations with successive powers of (R_c/R_ϵ), demonstrating numerical instability.

Section A: Question 2

Comment on how the simulation time-step size impacts the numerical solution, and which method is preferred in this case.

A.2.1 Simulation Parameters

I implemented all three discretization methods in MATLAB with the following circuit parameters: $V_{DC} = 10$ V, $V_0 = 0$ V, $C = 1\,\mu$F, and $h = 100\,\mu$, $T = 1ms$. I also simulated the Trapezoidal method in PSCAD with chatter attenuation disabled to allow direct comparison with the analytical solution.

A.2.2 Method Comparison and Selection

Forward Euler: Rejected

The Forward Euler method exhibits catastrophic numerical instability, as demonstrated in Figure 2. The current diverges exponentially with an amplification factor of $(R_c/R_\epsilon)^n \approx 10^{3n}$ at step n (Note that I've taken $R_\epsilon = R_c/1000$), reaching magnitudes exceeding 10^{17} A within microseconds. This non-physical behavior renders the method completely unsuitable for this circuit. Due to its extreme divergence, Forward Euler is shown separately and not included in the comparison plot with the other two methods.

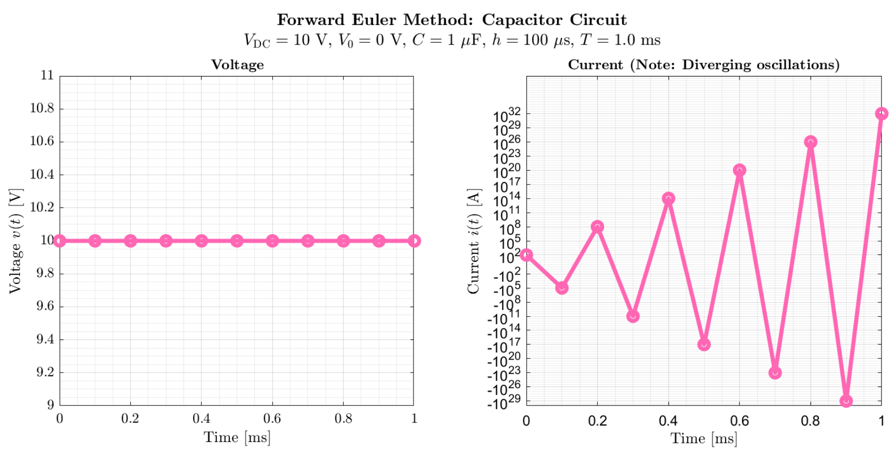

Figure 2: Forward Euler method showing exponential divergence in current magnitude (symmetric logarithmic scale). The oscillations grow with successive powers of (R_c/R_ϵ), demonstrating fundamental numerical instability.

Backward Euler vs. Trapezoidal

Figure 3 compares the Backward Euler and Trapezoidal methods against the PSCAD reference solution.

Key observations:

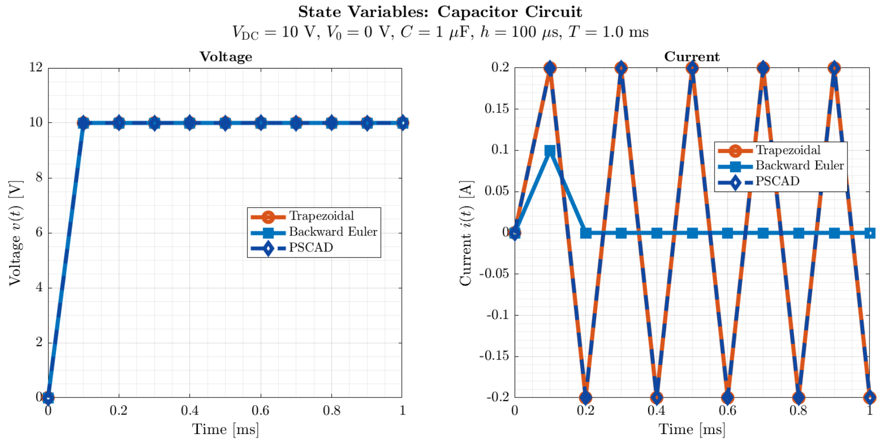

Figure 3: Comparison of Backward Euler, Trapezoidal, and PSCAD solutions. BE provides maximum damping (minimizing current oscillations), while Trapezoidal matches PSCAD exactly and exhibits numerical chatter characteristic of lossless circuit discretization.

- **Backward Euler**: Provides maximum numerical damping, driving $i(t) \to 0$ after the initial transient. This represents the *closest approach to the true steady-state solution* (zero current after capacitor charges), effectively suppressing the numerical oscillations entirely.
- **Trapezoidal**: Exhibits sustained bounded oscillations ("chatter") due to its energy-preserving nature. My MATLAB implementation and PSCAD solution (with chatter attenuation disabled) are *identical*, confirming correct implementation. The chatter is a numerical artifact of discretizing a lossless system.

A.2.3 Impact of Step Size h

The discretization resistance R_C scales linearly with step size h, directly coupling time-step selection to numerical behavior. Examining the step-by-step solutions in Tables 1 to 3, the impact of increasing h is summarized in Table 4:

Method	R_C	Chatter	Remark
BE	h/C	None	$i(t) = 0$ (damped) as $t \geq 2h$
Trapz	$h/(2C)$	$\downarrow$	Bounded oscillations persist
FE	h/C	Unstable	Amplification: $(R_c/R_\epsilon)^n$

Table 4: Impact of increasing h on discretization behavior (chatter magnitude $\downarrow$ = desirable).

A.2.4 Preferred Method

Backward Euler is preferred for this specific problem because it most closely represents the true steady-state behavior (zero current after capacitor fully charges), effectively eliminating the non-physical chatter artifacts. While in general Trapezoidal method offers higher accuracy (second-order vs. first-order), in this instance damping is prioritized over accuracy for the conclusion.

Section B (Basic Circuit Numerical Simulation): Question 1

The parameters of the circuit below are: $R = 10\Omega$, $L = 20\text{mH}$, $e(t) = 10\text{V}$ dc, and the switch is closed at $t = 0$. The circuit below is to be solved from $t = 0$ to $t = 10\text{ms}$ using: a) PSCAD software, b) your own MATLAB code based on the nodal analysis method.

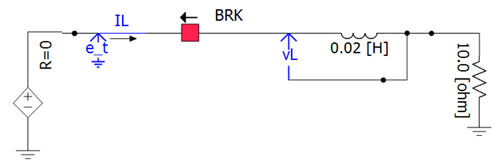

Figure 4: RL circuit with DC voltage source and switch

1) Obtain the solution for the inductor current and voltage using PSCAD with time-step sizes $\Delta t = 0.1\text{ms}$ and $\Delta t = 0.8\text{ms}$.

Section B : Question 2

2) Discretize the circuit and write a MATLAB code to solve the circuit step-by-step using the two discretization rules: a) Backward Euler, b) Trapezoidal, each with time-step sizes $\Delta t = 0.1\text{ms}$ and $\Delta t = 0.8\text{ms}$.

Section B: Question 3

3) Show the results for $i(t)$ obtained in the six cases (PSCAD, your Backward Euler code, and your Trapezoidal code; each with two different step sizes) superimposed in a (big enough) Plot. Show the same for $v_L(t)$ in a separate plot. Comment on the accuracy of the methods, impact of the time-step size, and how the PSCAD results compare with your Trapezoidal code.

Section C (Circuit Interruption): Question 1

The case below (figure on the left) represents a typical short-circuit fault in transmission lines and the breaker operation for line interruption. The equivalent circuit is shown in the figure on the right.

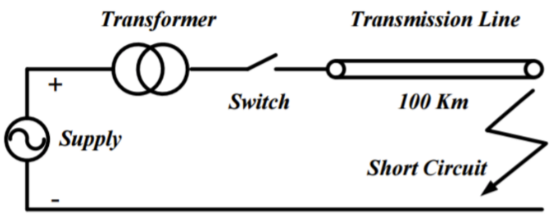

Figure 5: Transmission line with short-circuit fault

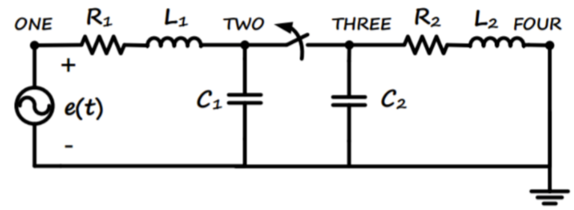

Figure 6: Equivalent circuit representation

On the left-hand side of the switch/breaker, we have the source voltage and a resistance and inductance representing the supply system combined with the transformer. The capacitor to the left of the breaker represents the combined substation capacitance due to the substation busbars. On the right-hand side of the breaker, we have the transmission line parameters: resistance, inductance, and capacitance represented as simple lumped parameters, where the right side is short-circuited due to the fault, assuming that the fault has already been applied. We have assumed a single-phase circuit here for simplicity. The parameters of the circuit below are: $e(t) = 230 \times \cos(\omega t)$ kV, $\omega = 377$ rad/s, $R_1 = 3\Omega$, $L_1 = 350$ mH, $R_2 = 50\Omega$, $L_2 = 100$ mH, $C_1 = 10$ nF, $C_2 = 600$ nF. Assume that the system is operating with zero initial conditions and the breaker is initially closed. At $t = 8$ ms, the breaker is commanded to open the line, but it will not actually open until the current passes through zero. The study period is from $t = 0$ to $t = 20$ ms.

1) Choose an appropriate simulation time-step size Δt (assuming the Trapezoidal integration rule) according to the time constants in the circuit. For this, evaluate the approximate resonant frequencies analytically before and after the breaker operation.

Section C: Question 2

2) Write your own MATLAB code to solve the system and also obtain the solution using PSCAD.

Section C: Question 3

3) Show the results for voltages at nodes Two and $Three$ and the fault current (i.e., from node $Four$ to $ground$) in three separate (big enough) plots. In each plot, superimpose the results from your code and also from PSCAD, both obtained using the same simulation time-step size obtained in part 1) for comparison. Comment on the accuracy of your results compared to those of PSCAD.

I Appendix: Derivation of Trapezoidal Discretization for Capacitor

To derive the equations v_t and i_t, the following steps are involved:

- Start with the governing equation for a capacitor:

$$i(t) = C\frac{dv(t)}{dt} \tag{7}$$

- Apply the trapezoidal rule to discretize the relationship between voltage and current:

$$v(t) - v(t - \Delta t) = \frac{\Delta t}{2C}\left[i(t) + i(t - \Delta t)\right] \tag{8}$$

- Define the equivalent resistance R_C and history terms:

$$R_C = \frac{\Delta t}{2C} \tag{9}$$

- Derive the Thevenin (series) and Norton (parallel) companion models.

The final equations for the capacitor are:

$$\boxed{v(t) = e_h + R_C i(t)} \tag{10}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{11}$$

where:

- $e_h = v(t - \Delta t) + R_C i(t - \Delta t)$
- $i_h = \frac{v(t-\Delta t)}{R_C} - i(t - \Delta t)$

II Appendix: Derivation of Backward-Euler Discretization for Capacitor

Starting from the capacitor constitutive relation

$$i(t) = C\frac{dv(t)}{dt}, \tag{12}$$

the Backward-Euler (BE) discretization at time t approximates the derivative by

$$\frac{dv(t)}{dt} \approx \frac{v(t) - v(t-\Delta t)}{\Delta t}. \tag{13}$$

Substituting and rearranging yields

$$i(t) \approx C\frac{v(t) - v(t-\Delta t)}{\Delta t}. \tag{14}$$

Writing the BE companion in a Thévenin/Norton form gives an equivalent resistance and history terms. The final boxed companion equations used in the main text (capacitor, BE) are

$$\boxed{v(t) = e_h + R_C\, i(t)} \tag{15}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{16}$$

with the BE companion parameters (for the capacitor):

$$R_C = \frac{\Delta t}{C}, \tag{17}$$

$$e_h = v(t-\Delta t), \tag{18}$$

$$i_h = \frac{v(t-\Delta t)}{R_C}. \tag{19}$$

These expressions are used when implementing the Backward-Euler discretization in the numerical examples and the BE table in the main text.

III Appendix: Derivation of Forward-Euler Discretization for Capacitor

Starting from the capacitor constitutive relation

$$i(t) = C\frac{dv(t)}{dt}, \tag{20}$$

the Forward-Euler (FE) discretization at time t approximates the derivative using the previous time step:

$$\frac{dv(t)}{dt} \approx \frac{v(t) - v(t-h)}{h}. \tag{21}$$

Substituting and rearranging yields

$$i(t) \approx C\frac{v(t) - v(t-h)}{h}. \tag{22}$$

To cast this in a companion model form similar to BE and trapezoidal methods, we introduce an artificial small resistance $R_\epsilon = h/C$ (as justified in the main text to avoid division by zero). The final boxed companion equations used in the main text (capacitor, FE) are

$$\boxed{v(t) = e_h(t)} \tag{23}$$

$$\boxed{i(t) = \frac{v(t) - e_h(t)}{R_\epsilon}} \tag{24}$$

with the FE companion parameters (for the capacitor):

$$R_c = \frac{h}{C} \quad , \tag{25}$$

$$e_h(t) = v(t-h) + R_c\, i(t-h), \tag{26}$$

$$i_h(t) = \frac{e_h(t)}{R_c}. \tag{27}$$

These expressions are used when implementing the Forward-Euler discretization in the numerical examples and the FE table in the main text.